

Spectral Non-Concentration as a Conditional Regularity Criterion for 3D Navier–Stokes on $\mathbb{T}^3$

Corrected framework version

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Abstract

We present a spectral framework for studying regularity of the three-dimensional incompressible Navier–Stokes equations on the torus $\mathbb{T}^3$. The central object is a shell-helical GCD interaction operator $H_N[a(t)]$ built from the normalized dyadic shell energy distribution of a Leray–Hopf solution. The main proved reduction is conditional: if the evolving operator remains within a fixed perturbative distance δ_0 of a frozen reference operator $\hat{H}_N^\mu$, then Weyl’s perturbation inequality gives a dynamic lower spectral gap

$$\inf_{t \geq 0} \lambda_{\min}(H_N[a(t)]) > -\frac{1}{2}.$$

The proof of this implication is elementary once the frozen gap input and finite-dimensional operator-continuity estimate are available. The unresolved step is the *SND Simplex Stability Lemma*: a quantitative dynamical statement asserting that the Navier–Stokes flow keeps the normalized shell vector $a(t)$ sufficiently close to the reference distribution μ , uniformly in time. This corrected version removes the contradictory claim that this step is closed. Section 7 records a proposed Gronwall closure mechanism, but treats it as a program for further proof rather than as an established theorem.

Status of claims in this corrected version

Claim	Status	Comment
Theorem 4.1: quantitative SND implies dynamic spectral gap	Proved conditional implication	Uses Weyl perturbation once hypotheses are assumed.
Lemma 3.1: finite-dimensional operator continuity	Proved	Direct triangle inequality.
Frozen reference gap $\lambda_{\min}(\hat{H}_N^\mu) > -1/2 + \delta_0$	Cited input from Paper 1	Full use for all N requires the cited proof to supply uniform constants, not only numerics.
Product condition $C_N \eta_N < \delta_0$	Numerical/conditional	The sample value $C_N \eta_N \approx 0.067 < 0.20$ is evidence at tested scales.
Lemma 6.1: SND Simplex Stability	Open	This is the remaining mathematical debt.
T1: analytic bound for C_N	Open	Expected route: GCD/Dirichlet-series estimates.
T2: quantitative cascade/Gronwall closure	Open	Section 7 gives the intended mechanism, not a completed proof.
Unconditional 3D Navier–Stokes global regularity	Not claimed here	Would follow only after Lemma 6.1 and the regularity criterion are fully established.

1 Introduction

The question of global regularity for the three-dimensional incompressible Navier–Stokes equations on $\mathbb{T}^3$ remains one of the central open problems in mathematical physics. We use the standard formulation

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0, \quad (1)$$

with initial data $u_0 \in H^1(\mathbb{T}^3)$.

This paper gives a conditional spectral reduction of the regularity problem to a quantitative statement about shell energy distribution. It does not claim that the full Navier–Stokes regularity problem is solved. The point of the paper is to isolate one precise closure target: a uniform-in-time simplex-stability estimate for the normalized shell energy vector.

1.1 Main claim structure

The logical chain is:

$$\text{SND Simplex Stability} \implies \text{operator perturbation bound} \implies \text{dynamic spectral gap}.$$

The final implication is proved in Theorem 4.1. The first implication is the open Lemma 6.1.

1.2 The central operator

For a Leray–Hopf solution $u(t)$, let P_j be a dyadic Littlewood–Paley shell projection and define shell energies

$$E_j(t) = \|P_j u(t)\|_{L^2}^2.$$

For a fixed cutoff N , define the normalized shell distribution

$$a_j(t) = \frac{E_j(t)}{\sum_{k \leq N} E_k(t)}, \quad a(t) \in \Delta_{N-1}, \quad (2)$$

whenever the denominator is nonzero. If the denominator vanishes, the solution is zero on the resolved shells and the operator can be defined by a harmless regularization or by excluding the trivial resolved state.

The shell-helical GCD interaction operator is

$$H_N[a] = \sum_{j=1}^N a_j B_j, \quad (3)$$

where each B_j is a symmetric matrix encoding GCD spectral coupling at shell j . The reference operator is

$$\hat{H}_N^\mu = H_N[\mu], \quad \mu_j = \frac{1}{N}.$$

Remark 1.1 (Boundedness versus smallness). *The Leray energy inequality*

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds \leq \|u_0\|_{L^2}^2$$

gives boundedness. It does not, by itself, control the shape of the shell-energy distribution. The normalization above places $a(t)$ in the compact simplex, but compactness is not proximity to μ . The missing estimate is precisely the quantitative SND/simplex-stability bound.

2 The GCD interaction operator

Let $\mathbf{m}$ denote the Moebius function. The unnormalized decorated GCD operator on $\{1, \dots, N\}$ is

$$Q_N(i, k) = \frac{\mathbf{m}(i/g)\mathbf{m}(k/g)g}{\sqrt{ik}}, \quad g = \gcd(i, k). \quad (4)$$

Define the degree weights

$$d_i = \sum_{k=1}^N |Q_N(i, k)|$$

and the normalized matrix

$$\hat{H}_N(i, k) = \frac{Q_N(i, k)}{\sqrt{d_i d_k}}. \quad (5)$$

The shell basis matrices are

$$(B_j)_{ik} = \hat{H}_N(i, k) \mathbf{1}\{\lfloor \log_2 \max(i, k) \rfloor = j\}. \quad (6)$$

2.1 Frozen reference input

The previous companion paper is cited for the frozen reference estimate. In a form sufficient for the present reduction, the required input is:

Hypothesis 2.1 (Frozen reference gap). *There exists $\delta_0 > 0$ such that, uniformly in the range under consideration,*

$$\lambda_{\min}(\hat{H}_N^\mu) > -\frac{1}{2} + \delta_0. \quad (7)$$

The working value used in the numerical examples is $\delta_0 = 0.20$.

Remark 2.2. *The original draft stated the finite computation $N \leq 800$ and an asymptotic limit in the same theorem. For an unconditional infinite-dimensional application, the asymptotic statement must be supplied by an analytic proof with constants uniform in N . In this corrected paper we treat the frozen gap as a cited input whose proof belongs to Paper 1.*

3 Operator continuity

Lemma 3.1 (Finite-dimensional operator continuity). *Let $H_N[a] = \sum_{j=1}^N a_j B_j$. For any $a, b \in \Delta_{N-1}$,*

$$\|H_N[a] - H_N[b]\|_{op} \leq C_N \|a - b\|_{\ell^1}, \quad C_N = \max_{1 \leq j \leq N} \|B_j\|_{op}. \quad (8)$$

Proof.

$$H_N[a] - H_N[b] = \sum_j (a_j - b_j) B_j.$$

By the triangle inequality,

$$\|H_N[a] - H_N[b]\|_{op} \leq \sum_j |a_j - b_j| \|B_j\|_{op} \leq \left(\max_j \|B_j\|_{op} \right) \|a - b\|_{\ell^1}.$$

□

Remark 3.2. *The original draft records numerical growth consistent with $C_N = O(\log N)$ at tested sizes. The analytic N -dependence remains part of the closure program.*

4 The master theorem

Theorem 4.1 (Quantitative SND implies dynamic spectral gap). *Assume the frozen gap Hypothesis 2.1. Assume also the quantitative SND perturbation bound*

$$\|H_N[a(t)] - \hat{H}_N^\mu\|_{op} < \delta_0 \quad \text{for all } t \geq 0. \quad (9)$$

Then

$$\inf_{t \geq 0} \lambda_{\min}(H_N[a(t)]) > -\frac{1}{2}. \quad (10)$$

Proof. Weyl's eigenvalue perturbation inequality gives

$$\lambda_{\min}(H_N[a(t)]) \geq \lambda_{\min}(\hat{H}_N^\mu) - \|H_N[a(t)] - \hat{H}_N^\mu\|_{op}.$$

Using the frozen gap and (9),

$$\lambda_{\min}(H_N[a(t)]) > \left(-\frac{1}{2} + \delta_0 \right) - \delta_0 = -\frac{1}{2}.$$

□

Corollary 4.2 (Simplex version). *If*

$$C_N \|a(t) - \mu\|_{\ell^1} < \delta_0 \quad \text{for all } t \geq 0, \quad (11)$$

then the conclusion of Theorem 4.1 holds.

Proof. Apply Lemma 3.1 with $b = \mu$. □

5 The quantitative SND condition

The working quantitative condition is

$$\|a(t) - \mu\|_{\ell^1} \leq \eta_N, \quad C_N \eta_N < \delta_0, \quad t \geq 0. \quad (12)$$

Numerically, the original draft used $C_N \approx 3.6$, $\eta_N \approx 0.039$, and hence $C_N \eta_N \approx 0.067 < 0.20$. This is a useful target value, but it is not a theorem until C_N and η_N are controlled analytically in the required range.

6 The remaining mathematical debt

Lemma 6.1 (SND Simplex Stability – open). *Let $a(t) \in \Delta_{N-1}$ be the normalized shell-energy distribution of a Leray–Hopf solution on $\mathbb{T}^3$, and let $\mu \in \Delta_{N-1}$ be the frozen reference distribution. Suppose SND is made quantitative so that the flow has no persistent coherent concentration in any dyadic shell or helicity sector. Then*

$$\|a(t) - \mu\|_{\ell^1} \leq \eta_N \quad \text{for all } t \geq 0, \quad (13)$$

with $C_N \eta_N < \delta_0$.

Status. Lemma 6.1 is open in this corrected version.

6.1 Two tasks required for closure

T1. Analytic bound for C_N . Prove an explicit estimate such as $C_N = O(1)$, or $C_N \leq A \log N + B$, and pair it with an η_N bound satisfying $C_N \eta_N < \delta_0$ uniformly.

T2. Dynamical shell-equidistribution estimate. Prove that the Navier–Stokes cascade dynamics, under a quantitative SND hypothesis, force the normalized shell vector to remain within the required simplex distance of the reference distribution. This must be obtained from shell-energy evolution, triad-transfer estimates, viscosity, and a quantitative flux condition. Leray boundedness alone is not enough.

7 Proposed Gronwall closure mechanism for T2

This section records the intended closure mechanism. It replaces the original draft’s contradictory heading “T2 – Closed Gronwall Proof.” The estimates below should be read as a proof plan until each constant is derived with the required uniformity in N and t .

Let

$$d(t) = \|a(t) - \mu\|_{\ell^1}.$$

The target inequality is

$$d'(t) \leq -\kappa d(t) + \varepsilon_N(t), \quad (14)$$

where $\kappa > 0$ is a dissipative/transfer rate and ε_N is a forcing term controlled by the quantitative SND condition.

A sufficient closure would be:

$$\sup_{t \geq 0} \varepsilon_N(t) < \kappa \eta_N \quad (15)$$

and an initial or entrance-time condition compatible with $d(t) \leq \eta_N$. Under these hypotheses, Gronwall gives

$$d(t) \leq e^{-\kappa(t-t_0)} d(t_0) + \frac{\sup_s \varepsilon_N(s)}{\kappa}. \quad (16)$$

This yields $d(t) \leq \eta_N$ after an entrance time, or for all time if the initial data already satisfies the compatible smallness condition. A separate argument is needed to cover the transient interval $[0, t_0]$ without assuming the conclusion.

The technical objective is therefore to prove a shell-flux estimate of the form

$$|\Phi_j(t) - \Phi_j^\mu(t)| \leq K \left(|a_j - \mu_j| + \sum_{|m-j| \leq 1} |a_m - \mu_m| \right) M(t), \quad (17)$$

with $M(t)$ controlled in a way that does not assume the desired regularity conclusion. If (17), (14), and (15) are proved with uniform constants, then Lemma 6.1 becomes a theorem.

Remark 7.1 (Why this is not yet an unconditional proof). *The closure mechanism requires three nontrivial steps: a quantitative triad-transfer bound, a positive uniform value of κ , and a forcing bound $\varepsilon_N < \kappa \eta_N$ independent of hidden smoothness assumptions. Until those estimates are supplied, this section is a roadmap, not a completed proof.*

8 Summary and outlook

The paper proves the perturbative spectral reduction:

$$\text{quantitative SND} \implies \text{dynamic spectral gap above } -\frac{1}{2}.$$

The exact remaining target is Lemma 6.1. If the analytic C_N estimate and the quantitative cascade/Gronwall closure are both established, then the conditional framework becomes a candidate route to a full regularity theorem. Until then, the mathematically safe claim is that this paper supplies a conditional criterion and a precise closure program.

References

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